MID 1
Subject: MCN

Q1. Define model in network computing. List different network models.

A model in network computing is a simple structure that shows how computers communicate and share data in a network.

Different network models:

1.Centralized Computing
2.Distributed Computing
3.Collaborative Computing
4.Client-Server Model
5.Peer-to-Peer Model

Q2. Draw and explain OSI Model. Write responsibilities of each layer.

OSI Model (7 Layers):

7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

Explanation & Responsibilities:

1.Physical Layer – Sends raw bits through cables or wireless signals.
It defines voltage, cables, connectors, and data transmission speed.

2.Data Link Layer – Provides error detection and controls data frames. Uses MAC address.
It ensures data is sent correctly between two directly connected devices.

3.Network Layer – Handles IP addressing and routing of data.
It finds the best path to send data from source to destination.

4.Transport Layer – Provides end-to-end delivery and error control (TCP/UDP).
It breaks data into segments and reassembles them at destination.

5.Session Layer – Establishes, manages, and terminates sessions.

It controls dialogue between sender and receiver.

6.Presentation Layer – Converts data format and provides encryption.
It compresses data to reduce size before transmission.

7.Application Layer – Provides services like email, web browsing, and file transfer.
It allows users to interact with network applications.

Q3. Draw and explain TCP/IP Model. Write responsibilities of any two layers.

TCP/IP Model

4. Application Layer
3. Transport Layer
2. Internet Layer
1. Network Access Layer

Explanation of Layers:

1.Network Access Layer

It handles physical transmission of data.
It sends data through cables or wireless network.

2.Internet Layer

It uses IP address for communication.
It is responsible for routing data from source to destination.

3.Transport Layer

It provides end-to-end communication.
It ensures reliable data transfer using TCP or fast transfer using UDP.

4.Application Layer

It provides services like email, web browsing, and file transfer.
It allows users to access network applications.

Responsibilities of Any Two Layers:

1.Transport Layer:

Provides reliable data transfer (TCP).

Performs error control and flow control.

2.Internet Layer:

Provides logical addressing (IP address).
Selects best path for data routing.

Q4. Write a short note on Traffic Profile and Congestion Control.

1. Traffic Profile

A traffic profile describes the nature and behavior of data transmission in a network.
It defines the data rate (how fast data is sent).
It defines burst size (amount of data sent at one time).
It defines bandwidth requirement.
It helps in improving network performance and planning.
Traffic profile is important to manage network resources properly and avoid overload.

2. Congestion Control

Congestion occurs when too many devices send data at the same time and the network becomes slow.
Congestion control is the technique used to control traffic and maintain smooth communication.

If congestion is not controlled:

1.Packets may be lost
2.Delay increases
3.Network performance decreases

Types of Congestion Control:

1.Open-Loop Congestion Control

Prevents congestion before it occurs.
No feedback is used.
Example: Proper network design and traffic shaping.

2.Closed-Loop Congestion Control

Detects congestion after it happens.
Uses feedback from network.
Example: TCP reduces sending rate when congestion is detected.

Q5. Compare TCP/IP Model and OSI Model.

1) OSI Model

1.It has 7 layers.
2.Developed by ISO (International Organization for Standardization).
3.It is a theoretical model.
4.Layers are strictly separated.
5.Session and Presentation are separate layers.
6.Mainly used for learning and understanding networking.

2) TCP/IP Model

1.It has 4 layers.
2.Developed by DARPA.
3.It is a practical model used on the Internet.
4.Layers are not strictly separated.
5 Session and Presentation are included in Application layer.
6.Used in real-world communication (Internet).

Q6. Compare IPv4 and IPv6

IPv4

1.Uses 32-bit address.
2.Limited number of IP addresses.
3.Address written in decimal format (e.g., 192.168.1.1).
4.Uses NAT.
5.Header size is 20 bytes.
6.Less security (IPSec optional).

IPv6

1.Uses 128-bit address.
2.Very large number of IP addresses.
3.Address written in hexadecimal format (e.g., 2001:db8::1).
4.No need of NAT.
5.Header size is 40 bytes.
6.Better security (IPSec built-in).

Q7. Explain Subnetting and Supernetting.

1) Subnetting

Subnetting is the process of dividing one large network into smaller networks called subnets.
It improves network performance.
It reduces network traffic.
It provides better security and management.
It uses subnet mask to divide the network.

Example:
Suppose we have Class C network: 192.168.1.0 with default mask 255.255.255.0
If we change subnet mask to 255.255.255.192,

we create 4 subnets:
- 192.168.1.0
- 192.168.1.64
- 192.168.1.128
- 192.168.1.192

Each subnet will have limited hosts. This is subnetting.

2) Supernetting

Supernetting is the process of combining multiple small networks into one large network.
It reduces routing table size.
It improves routing efficiency.
It is also called CIDR (Classless Inter-Domain Routing).
It combines multiple IP networks into one.

Example:
If an ISP combines:
- 192.168.0.0
- 192.168.1.0
- 192.168.2.0
- 192.168.3.0

into a single larger block, this process is called supernetting.
Thus, subnetting divides networks, while supernetting combines networks.

Q8. Explain Network Address Translation (NAT).Write its Advantages and Disadvantages.

Network Address Translation (NAT)

Network Address Translation (NAT) is a technique used in networking to convert private IP addresses into a public IP address when devices access the Internet.
Private IP addresses are used inside a local network, but they cannot be used directly on the Internet. NAT helps in translating these addresses.

Working of NAT:

A device in a private network sends a request to the Internet.
The router changes the private IP address into a public IP address.
The data is sent to the Internet using the public IP.
When the reply comes back, NAT converts the public IP back to the correct private IP.

Types of NAT:

1.Static NAT – One private IP is mapped to one public IP.
2.Dynamic NAT – Private IP is mapped to any available public IP.
3.PAT (Port Address Translation) – Many private IPs share one public IP using port numbers.

Advantages of NAT:

1.Saves IPv4 addresses.
2.Provides basic security by hiding private IP addresses.
3.Allows multiple devices to use one public IP.

Disadvantages of NAT:

1.Reduces network speed.
2.Configuration can be complex.
3.Some applications may face connectivity problems.

Q9. Explain IPv4 Addressing Scheme.

Introduction:
IPv4 (Internet Protocol version 4) is a logical addressing system used to identify devices in a network.
It uses a 32-bit address.
An IPv4 address is written in decimal format and divided into 4 parts (octets).

Example: 192.168.1.1
Each part ranges from 0 to 255.

Structure of IPv4 Address:

Total bits: 32 bits
Divided into 4 octets (8 bits each)
Format: X.X.X.X

It has two parts:

1.Network ID – Identifies the network

2.Host ID – Identifies the device in that network

Classful Addressing:

IPv4 has 5 classes:

Class A – Large networks (1 – 126)
Class B – Medium networks (128 – 191)
Class C – Small networks (192 – 223)
Class D – Multicast
Class E – Experimental

Classless Addressing (CIDR):

Does not depend on classes.
Uses subnet mask like /24, /16 etc.
More efficient use of IP addresses.

Example: 192.168.1.0/24

Q10. Build Subnet Mask for IP Address 19.168.10.15

Step 1: Identify the Class of IP Address
Given IP Address: 19.168.10.15
The first octet is 19.
IP range for Class A is 1 to 126.
Since 19 comes in this range, it is a Class A IP address.

Step 2: Default Subnet Mask of Class A
The default subnet mask for Class A is:
255.0.0.0

In binary form:
11111111.00000000.00000000.00000000

Step 3: Network and Host Portion
In Class A:
First octet (19) → Network ID
Last three octets (168.10.15) → Host ID

So,
Network ID = 19.0.0.0

Host ID = 168.10.15

Final Answer:
The IP address 19.168.10.15 belongs to Class A,

and its default subnet mask is:
255.0.0.0

Q11. Compare WWW and Internet

Internet

1.It is a network of networks connected worldwide.
2.It is the main infrastructure for communication.
3.It uses TCP/IP protocol.
4.It provides services like email, FTP, and WWW.
5.It connects millions of computers globally.

WWW (World Wide Web)

1.It is a collection of web pages and websites.
2.It works on the Internet.
3.It uses HTTP/HTTPS protocol.
4.It is a service provided by the Internet.
5.It is accessed using web browsers like Chrome.

Q12. Explain IPv6

IPv6 (Internet Protocol version 6) is the new version of Internet Protocol used to identify devices on a network.

Features of IPv6:

It uses 128-bit address, so it provides a very large number of IP addresses.
It solves the problem of IPv4 address shortage.
It provides better security (IPSec support).
It improves routing and network performance.

Example of IPv6 address:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv6 is used for modern networks and future Internet growth.

Q13. Define the term Mobile Computing. Explain in brief.

Definition:
Mobile Computing is a technology that allows users to access data and communicate using mobile devices without a fixed wired connection.
It enables communication anytime and anywhere.

Explanation (Brief):
It uses devices like smartphones, tablets, and laptops.
It works through wireless networks like Wi-Fi and mobile data.
It allows users to access the Internet, email, and applications while moving.
It supports real-time communication and data sharing.

Example:
Using a smartphone to browse the Internet or send email while traveling.

Q14. Explain Mobile Computing Architecture.

Introduction:

Mobile Computing Architecture shows how mobile devices communicate with servers through wireless networks.
It explains the structure of mobile communication system.

Main Components:

1.Mobile Device

Devices like smartphone, tablet, or laptop.
Used by users to access services.

2.Base Station (Wireless Network)
Provides wireless connection.
Connects mobile device to network.

3.Network Infrastructure
Includes routers, switches, and communication links.
Transfers data between device and server.

4.Server / Internet
Stores data and applications.
Provides services like email, web, database access.

Working:

Mobile device → Base station → Network → Server → Response back to device.

Q15. Evolution of Mobile Computing

Evolution of Mobile Computing means the development of mobile communication technology over time.

1.1G (First Generation)
Only voice communication.
Analog signals used.
Low quality and less security.

2.2G (Second Generation)
Digital communication started.
SMS and basic data services available.
Better voice quality.

3.3G (Third Generation)
Internet and video calling supported.
Faster data speed.

4.4G (Fourth Generation)
High-speed Internet.
HD video streaming and online gaming.

5.5G (Fifth Generation)
Very high speed and low delay.
Supports IoT and smart devices.